

Cal. 15

Tarea 27

1.- Derivar las siguientes funciones

a) $2x^5 - x = 3y^3 + 2y + 8$

$$10x^4 - 1 = 9y^2 y' + 2y'$$

$$10x^4 - 1 = (9y^2 + 2)y'$$

$$y' = \frac{10x^4 - 1}{9y^2 + 2}$$

b) $2x^4 y^2 - xy^3 + 8y = 4\sqrt{x}$

$$2(x^4 2yy' + y^2 4x^3) - (x 3y^2 y' + y^3) + 8y' = 4 \frac{1}{2\sqrt{x}}$$

$$4x^4 yy' + 8x^3 y^2 - 3xy^2 y' - y^3 + 8y' = \frac{2}{\sqrt{x}}$$

$$4x^4 yy' - 3xy^2 y' + 8y' = y^3 - 8x^3 y^2 + \frac{2}{\sqrt{x}}$$

$$(4x^4 y - 3xy^2 + 8)y' = y^3 - 8x^3 y^2 + \frac{2}{\sqrt{x}}$$

$$y' = \frac{y^3 - 8x^3 y^2 + \frac{2}{\sqrt{x}}}{4x^4 y - 3xy^2 + 8}$$

c) $\sin(xy) = x^2 + y$

$$\cos(xy)(xy' + y) = 2x + y'$$

$$(xy' + y)(\cos(xy)) = 2x + y'$$

$$xy' \cos(xy) + y \cos(xy) = 2x + y'$$

$$xy' \cos(xy) - y' = 2x - y \cos(xy)$$

$$[x \cos(xy) - 1]y' = 2x - y \cos(xy)$$

$$y' = \frac{2x - y \cos(xy)}{x \cos(xy) - 1}$$

Nombre:

Día

Mes

Año

Folio

Tema:

2. Obtener y' de la ecuación $x^3 - 3y^2 + y^3 = y$ y calcular su valor en el punto $P(2, -1)$

$$3x^2 - 6yy' + 3y^2y' = y'$$

$$-6yy' + 3y^2y' - y' = -3x^2$$

$$6yy' - 3y^2y' + y' = 3x^2$$

$$(6y - 3y^2 + 1)y' = 3x^2$$

$$y' = \frac{3x^2}{6y - 3y^2 + 1}$$

$$y' \bigg|_{P(2, -1)} = \frac{3(2)^2}{6(-1) - 3(-1)^2 + 1} = \frac{12}{-6 - 3 + 1} = \frac{12}{-8} = -\frac{3}{2}$$

$P(2, -1)$

3. Sea la función

Obtener su derivada

$$\begin{cases} x = 2t^2 - t \\ y = \sqrt{t} - 1 \end{cases}$$

a) Usando la Fórmula

b) Eliminando el parámetro "t" y derivando el resultado.

a)

$$D_x y = \frac{D_t y}{D_t x} = \frac{\frac{1}{2\sqrt{t}}}{4t - 1} = \frac{1}{2(4t - 1)\sqrt{t}}$$

b) Despejando "t" de a)

$$\sqrt{t} = y + 1$$

$$t = (y + 1)^2$$

Sust. t'' en ①

$$x = 2[(y+1)^2 - (y-1)^2]$$

$$x = 2(y+1)^2 - (y-1)^2 \dots ③$$

Derivando ③

$$1 = 2 \cdot 4(y+1)^3 y' - 2(y-1) y'$$

$$[2 \cdot 4(y+1)^3 - 2(y-1)] y' = 1$$

$$y' = \frac{1}{8(y+1)^3 - 2(y-1)}$$

Verif.

Sust t'' en ①

$$y' = \frac{1}{8(y+1)^3 - 2(y-1)}$$

$$y' = \frac{1}{8(y+1)^3 - 2(y-1)}$$

4.- Sea la función $\begin{cases} x = \operatorname{ang} \operatorname{tg} 2t^2 \\ y = t^3 \end{cases}$

Obtener $D_x y$ y valorarla para $t=1$

$$D_x y = \frac{D_x y}{D_x x} = \frac{\frac{3t^2}{4t^2}}{\frac{1 + (2t^2)^2}{1 + 4t^4}} = \frac{3t^2}{4t^2} \cdot \frac{1 + 4t^4}{1 + 4t^4}$$

$$D_x y \Big|_{t=1} = \frac{3[1+4(1)^4]}{4} = \frac{15}{4}$$

$$\frac{3(1+4t^4)}{4}$$